

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Performance Engineering, Systems for Data Science, Optimization in CS, Advanced Machine Learning, Advanced NLP

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, C++, Java, JavaScript, TypeScript, SQL, Bash / shell

Scripting & Tooling

Python scripting, Bash / shell, system interfaces, GTest / CMake, OpenMP / SIMD concepts, Linux profiling

Quality & Security

Automated & integration testing, CI/CD, code reviews, authentication hardening, VAPT remediation, secure-by-default patterns

Data & Reporting

Apache Spark / PySpark, Spark UI, data analysis, pipeline instrumentation, SQL, PostgreSQL, MongoDB

Cloud & DevOps

AWS (S3, KMS), GCP, Azure, Docker, Git, GitHub Actions, Linux

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Owned backend reliability and security for **multi-tenant NHS eTriage systems** (Node.js, MongoDB) serving **8+ UK clients** and processing **10,000+ monthly triage submissions** integrated with Mayden's iaptus platform; contributed to **\$340K+** in revenue.
- Monitored production logs during feature releases and client integrations, investigated incident bugs, and shipped fixes for tenant-specific and integration issues; improved operational stability through authentication hardening, rate limiting, and audit logging.
- Remediated **20+ VAPT findings** and added integration tests to protect sensitive clinical workflows against regressions; enforced secure-by-default patterns across tenant isolation and backend access paths.
- Joined a **code-quality** pilot using GitHub Copilot for tech-debt PRs, code migrations, and security-finding triage, and maintained a versioned conventions file to keep generated code aligned with team standards.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js + Fastify backed by **PostgreSQL**, with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.

Projects

Spark ETL Performance Optimization (NYC TLC Trip Data)

github.com/Deva-1903/Spark-ETL-Optimization

Apache Spark / PySpark, SQL, Python | 1.4B-row dataset

- Built a Python/Spark **data-analysis pipeline** over a **1.4B-row** dataset, tuning broadcast joins, partitioning, and adaptive settings to cut runtime **25%** while keeping the workflow process-oriented and repeatable.
- Used **Spark UI** stage-level metrics to localize shuffle and skew bottlenecks and to present a documented, reproducible analysis of where time was spent, a reporting workflow other jobs could reuse.

AutoEval - Reporting & Evaluation Tool

github.com/Deva-1903/AutoEval

Python, Evaluation, Git Automation

- Built a Python tool that scores systems against an evaluation suite and commits accepted updates back to Git with **auditable report cards**, with guardrails (schema checks, duplicate rejection, holdout protection) for a reproducible, process-oriented workflow.

AgenticSearch - Tested Python Service

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, OpenAI / Groq, SQLite

- Hardened a multi-stage Python service with **150+ automated tests**, provider routing/fallback, and pipeline instrumentation, prioritizing **code quality** and review-friendly debuggability.
- Structured the service as modular, testable stages (typed planning, extraction, verification) with clear boundaries, process-oriented tooling that keeps a codebase maintainable.

Underdogs Fitness - Full-Stack Platform & CI/CD Automation

underdogsfitness.in

Node.js, React, MongoDB | live in production 3+ years

- Built and solo-maintained a production platform for **3+ years** and set up **git-webhook auto-deploy** and cron-driven automation, dev-process tooling that cut the client's manual work by **~70%**.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**.
- Selected for **GirlScript Summer of Code (GSSoC) 2023**; contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.